This Zenodo deposit contains a publicly available description of the Dataset:

"Single-cell RNAseq of colonic lamina propria in wild type and PINK1 KO mice after C. rodentium infection".

Dataset Description:

We performed 10X Genomics single-cell RNA sequencing of colonic lamina propria cells from wild type and PINK1 KO mice following either 1-week or 2-weeks post C. rodentium infection. The cells were pooled from 3 mice per group of both sexes at 8-12 weeks of age. This dataset contains raw FASTQ files from mouse colonic lamina propria, sorted into 8 groups namely wild type and PINK1 KO uninfected and infected mice at 1- or 2-weeks post-infection. Sequencing was performed using NovaSeq 6000 S4 PE 100bp. Reads were processed using the 10X Genomics Cell Ranger Single Cell 2.0.0 pipeline. FASTQs generated from sequencing output were aligned to the mouse GRCm38 reference genome using STAR algorithm 2.7.3a.

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Dataset Name: desjardins-mouse-sc-rnaseq-colon-immune-pink1, vv0.1

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ASAP Lab:

ASAP Project: PINK1 deficiency rewires early immune responses in a mouse model of Parkinson's disease triggered by intestinal infection

Project Description: Parkinson's disease is characterized by a period of non-motor symptoms, including gastrointestinal dysfunction, preceding motor deficits by several years to decades. This long prodrome is suggestive of peripheral immunity involvement in the initiation of disease. We previously developed a model system in PINK1 KO mice displaying PD-like motor symptoms at late stages following intestinal infections. In this study, we map the initiating immune events at the site of infection in this model. Using single-cell RNAseq, we demonstrate that peripheral myeloid cells are the earliest highly dysregulated immune cell type followed by an aberrant T cell response shortly after. We also demonstrate an increased propensity for antigen presentation and that activated myeloid cells acquire a proinflammatory profile capable of inducing cytotoxic T cell responses. Together, our study provides the first evidence that PINK1 is a key regulator of immune functions in the gut underlying early PD-related disease mechanisms.

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